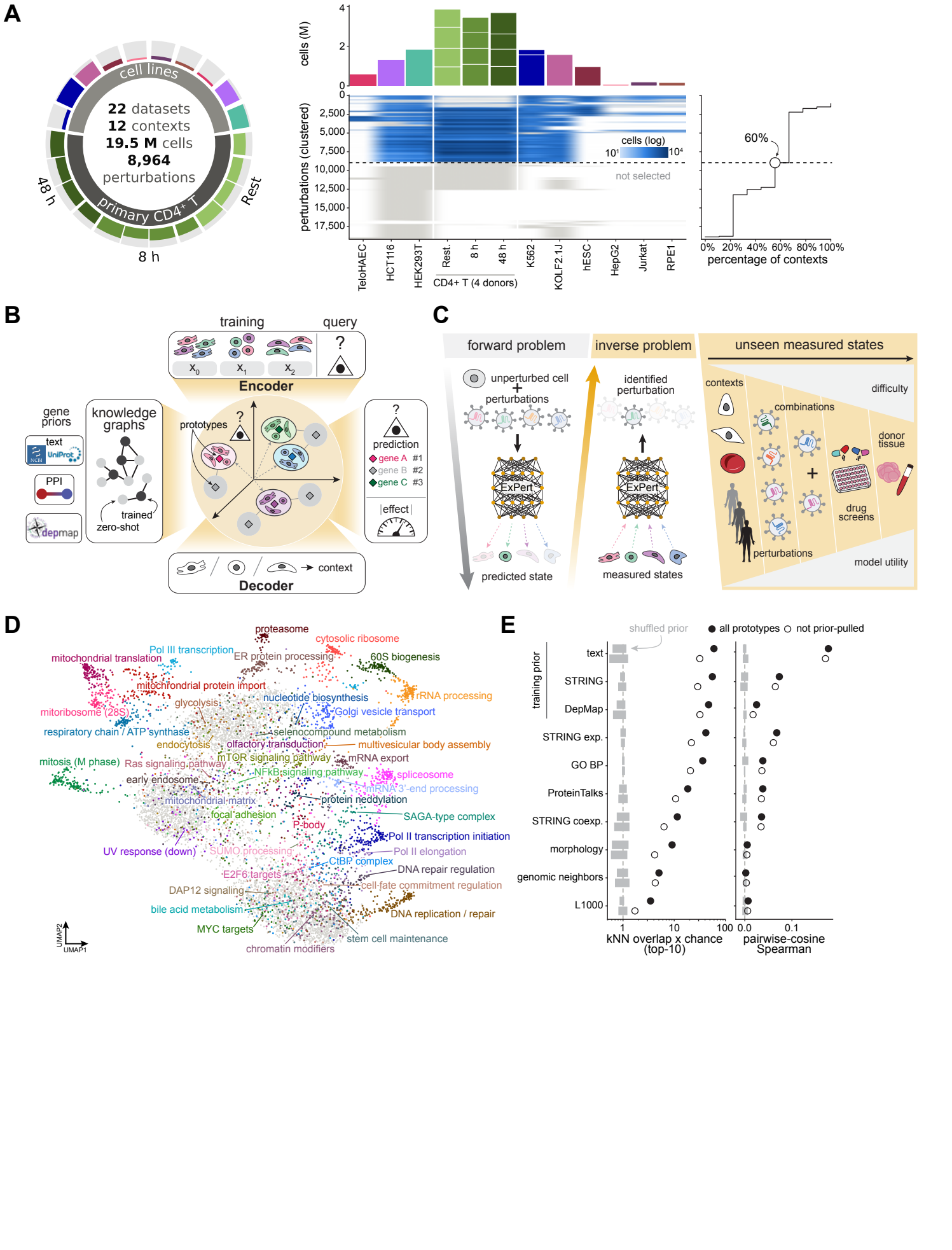
Figure 1


A
cell lines
22 datasets
12 contexts
19.5 M cells
8,964
perturbations
primary CD4+ T
48 h
8 h
Rest
cells (M)
perturbations (clustered)
cells (log)
not selected
60%
percentage of contexts
TeloHAEC
HCT116
HEK293T
Rest.
8 h
48 h
K562
KOLF2.1J
hESC
HepG2
Jurkat
RPE1
CD4+ T (4 donors)
B
training
query
Encoder
gene priors
knowledge graphs
text
PPI
depmap
trained
zero-shot
prototypes
prediction
gene A #1
gene B #2
gene C #3
effect
context
Decoder
C
forward problem
inverse problem
unseen measured states
unperturbed cell
perturbations
identified perturbation
ExPert
predicted state
measured states
contexts
difficulty
combinations
donor tissue
drug screens
perturbations
model utility
D
proteasome
cytosolic ribosome
Pol III transcription
mitochondrial translation
ER protein processing
60S biogenesis
mitochondrial protein import
rRNA processing
glycolysis
nucleotide biosynthesis
mitoribosome (28S)
Golgi vesicle transport
respiratory chain / ATP synthase
selenocompound metabolism
endocytosis
olfactory transduction
multivesicular body assembly
mitosis (M phase)
Ras signaling pathway
mTOR signaling pathway
mRNA export
NFkB signaling pathway
spliceosome
early endosome
mRNA 3'-end processing
mitochondrial matrix
protein neddylation
focal adhesion
SAGA-type complex
P-body
Pol II transcription initiation
UV response (down)
SUMO processing
Pol II elongation
CtBP complex
DNA repair regulation
E2F6 targets
cell-fate commitment regulation
DAP12 signaling
bile acid metabolism
DNA replication / repair
MYC targets
stem cell maintenance
chromatin modifiers
UMAP2
UMAP1
E
shuffled prior
all prototypes
not prior-pulled
training prior
text
STRING
DepMap
STRING exp.
GO BP
ProteinTalks
STRING coexp.
morphology
genomic neighbors
L1000
kNN overlap x chance (top-10)
pairwise-cosine Spearman